Exercise list R10 — Macroeconomic implications of corporate-finance frictions

Answer key (full derivations in `lista-r10-macro-implications-solutions.pdf`)

1. Investor protection and the interest rate (Tirole 13.1)

(i) IC $(\Delta p)R_b \geq B \Rightarrow$ pledgeable income $p_H(R - B/\Delta p)$; financing iff $(1+r)(I-A) \leq p_H(R-B/\Delta p)$, i.e. $A \geq A^*(r) = I - p_H(R - B/\Delta p)/(1+r)$. **(ii)** Savings = investment: $S(r) + \int_0^I A\, g(A)\, dA = [1 - G(A^*(r))]I$. Left side rises with r, right side falls ($\partial A^*/\partial r > 0$); $\partial A^*/\partial B > 0$, so $dr/dB < 0$: better protection (lower B) raises r. Example (G uniform on $[0,1]$, $S = 2r$, $p_H = 0.8$, $R = 1.5$, $\Delta p = 0.4$): $B = 0.3 \Rightarrow r = 0.038795$; $B = 0.2 \Rightarrow r = 0.110272$. Matches the book (pp. 616–617).

2. Time-consistent government policy (Tirole 16.2)

(i) τ cancels from the IC $\Rightarrow (p_H + \tau)(R - B/\Delta p)I = I - A$, $I(\tau) = A/[1 - (p_H + \tau)(R - B/\Delta p)]$. Ex post ($I$ sunk): $\max_\tau (p_H + \tau)RI - \gamma(\tau)I \Rightarrow \gamma'(\tau^*) = R$. **(ii)** $\max_\tau [(p_H + \tau)R - 1 - \gamma(\tau)]I(\tau)$ $\Rightarrow [\gamma'(\tau^c) - R]I = [(p_H + \tau^c)R - 1 - \gamma(\tau^c)]\, dI/d\tau > 0$, so $\tau^c > \tau^*$. **(iii)** Weight $\omega < 1$ on foreign investors: $\gamma'(\tau) = R - (1-\omega)(R - B/\Delta p) < R$, so $\tau < \tau^*$ and less lending. Example ($\gamma = 10\tau^2$, $p_H = 0.5$, $R = 2$, $B/\Delta p = 1$, $A = 1$): $\tau^* = 0.1$, $\tau^c = (10 - \sqrt{60})/20 = 0.112702$; $\omega = 0.5$: $\tau = 0.075$. Matches the book (p. 622).

3. Imperfectly correlated shocks (Tirole 15.3)

Efficient allocation $x = 1$, $I = A/(1 + \bar{\rho} - \rho_0)$ (independent of θ). Binding state (common shock high): need $[\theta + (1-\theta)\lambda](\rho_H - \rho_0)I$, supply $(1-\theta)(1-\lambda)(\rho_0 - \rho_L)I$. With the breakeven $(1-\lambda)(\rho_0 - \rho_L)I = (I - A) + \lambda(\rho_H - \rho_0)I$, supply $\geq$ need iff $(1-\theta)(I - A) \geq \theta(\rho_H - \rho_0)I$, i.e. $\theta \leq \theta^*$. The other state never binds (net borrowing). Example ($\lambda = 0.3$, $\rho_L = 0.2$, $\rho_0 = 0.6$, $\rho_H = 1$, $A = 1$): $I = 1.190476$, $\theta^* = 2/7 = 0.285714$. Matches the book (p. 621).

4. Property rights and international finance (Tirole Review Problem 5)

(i) Breakeven $(1-t)\rho_0 I = I - A \Rightarrow I^* = A/[1 - (1-t^*)\rho_0]$. Ex post: $\max_t [B_0(t) - \theta t]\rho_0 I \Rightarrow$ $B_0'(t^*) = \theta^* = S_D/(I^* - A)$; unique $t^* \in (0,1)$ because $S_D < \rho_0 A/(1-\rho_0)$. **(ii)** Higher $S_D \Rightarrow$ $t^* \downarrow$, $\theta^* \uparrow$, $I^* \uparrow$, and $U_b = [(\rho_1 - 1) - (t - B_0(t))\rho_0]I$ rises. **(iii)** Commitment: $t = 0$. **(iv)** Discrimination: foreigners fully taxed, domestic untaxed; no foreign lending, $I^* - A = S_D$, domestic investors earn $\rho_0(A + S_D) > S_D$. Example ($B_0 = \ln(1+t)$, $\rho_0 = 0.6$, $\rho_1 = 1.5$, $A = 1$, $S_D = 0.5$): $t^* = 1/3$, $\theta^* = 0.75$, $I^* = 1.666667$, $U_b = 0.787682$ (commitment: 1.25); $S_D = 0.6$: $t^* = 0.276984$, $U_b = 0.848673$. Matches the book (p. 635).

5. Borrowing abroad (Tirole 16.1)

(i) IC $p(RI - R_l) \geq \mu I$, breakeven $pR_l = I - A \Rightarrow I = A/[1-(pR-\mu)]$, $U_b = (pR-1)A/[1-(pR-\mu)]$: basic model with $p_H = p$, $p_L = 0$, $B = \mu$. **(ii)** Balanced tax/guarantee: same pledgeable income, same I. Public debt D: $I = (A-D)/[1-(pR-\mu)]$; consumption μI falls by $\mu D/[1-(pR-\mu)] > D$ (book: gain $(pR-1)(A-D)/[1-(pR-\mu)]$ over $A - D$). **(iii)** Commitment: μ_L; no commitment: μ_H ex post, $U_b = (pR-1)A/[1-(pR-\mu_H)]$. **(iv)** $e\mathcal{R} = pR_l$, $pR_l/e = I - A \Rightarrow I = A + \mathcal{R}$, $e = (1 + A/\mathcal{R})(pR - \mu)$ if ≥ 1 (else $e = 1$). Example ($p = 0.5$, $R = 3$, $\mu = 0.6$, $A = 1$): $I = 10$, $U_b = 5$; $D = 0.2$: $I = 8$, loss 1.2; $\mu_H = 0.8$: $U_b = 1.666667$; $\mathcal{R} = 2$: $e = 1.35$, $I = 3$. Matches the book (p. 622).

6. Shleifer–Vishny with decreasing returns (Tirole 14.1)

Price (seller's take-it-or-leave-it) = buyer's extra pledgeable income: $P_j I_j = p_H[R(I_i+I_j)-R(I_i)]-bI_j$, $b = p_H B/\Delta p$, decreasing in I_i. With $\pi_s = x(1-\mu) = (1-x)(1-\nu)$,

$$\frac{\partial U_{bj}}{\partial I_i} = \pi_s p_H\Big[\frac{B}{\Delta p} - (R'(I_i) - R'(I_i+I_j))\Big],$$

windfall (+) vs. lower resale price (−): negative when concavity is strong. I_i also lowers j's borrowing capacity. Example ($R = 2\sqrt{I}$, $x = 0.8$, $\nu = 0$, $p_H = 0.8$, $B/\Delta p = 0.5$): $+0.033137$ at $I = 1$, -0.013726 at $I = 0.25$. No answer in the book.

7. Too few asset reallocations (Tirole 14.4)

Resell iff $\rho_0 < c$, proceeds to investors; $\pi(c) = F(c)\hat{\rho}_0 + \int_c \rho_0 dF$, $I = A/(1-\pi)$, $U_b = [\pi + \Delta\rho(1-F) - 1]I$. FOC: $\rho_0^* = \hat{\rho}_0 - [1-\pi(\rho_0^*)]/[1-F(\rho_0^*)]$, so $\hat{\rho}_0 - \Delta\rho < \rho_0^* < \hat{\rho}_0$ ($\pi < 1$, NPV> 0): states $\rho_0 \in (\rho_0^*, \hat{\rho}_0)$ are not reallocated. Example ($\rho_0 \sim U[0,1]$, $\hat{\rho}_0 = 0.7$, $\Delta\rho = 0.5$): $\rho_0^* = 1 - \sqrt{0.6} = 0.225403$. Matches the book (p. 619), whose U_b display omits $-I$.

8. Inside liquidity (Tirole Review Problem 6)

(i) $(1+\lambda\rho x)I - A = (1-\lambda+\lambda x)\rho_0 I \Rightarrow I = A/\{(1+\lambda\rho x) - [1-\lambda+\lambda x]\rho_0\}$. **(ii)** $U_b = \frac{\rho_1 - c(x)}{c(x) - \rho_0}A$, $c(x) = \frac{1+\lambda\rho x}{1-\lambda+\lambda x}$, $c' \propto (1-\lambda)\rho - 1 \Rightarrow x = 1$ iff $(1-\lambda)\rho \le 1$. **(iii)** Shortage $(\rho - \rho_0)I$. **(iv)** Index worth $(1-\lambda)\rho_0 I$; insufficient iff $\rho - \rho_0 > (1-\lambda)\rho_0$, though aggregate liquidity suffices; pool via credit lines. Example ($\lambda = 0.4$, $\rho = 1.2$, $\rho_0 = 0.5$, $\rho_1 = 2$, $A = 1$): $I = 1.020408$, $U_b = 0.530612$, shortage 0.714286. Matches the book (p. 635).

9. Soft budget constraint (Tirole 16.4)

(i) NPV$(x) = \rho(y_H - w) + (1-\rho)x(y_L - w) - I$; $x^* = 1$ (higher NPV, financeable by assumption). Market refinances $y_L - \mathcal{R} < w$: liquidity $L^* = w + \mathcal{R} - y_L$. **(ii)** Ex post: continue iff $y_L - \lambda s \ge -\lambda w_u$; with $s = w + \mathcal{R} - y_L$ this is the assumption, so the government always bails out $\Rightarrow L = 0$, $s = w + \mathcal{R} - y_L$. With commitment: $s = 0$, firms hoard L^*. Example ($y_L = 1$, $w = 0.8$, $\mathcal{R} = 0.5$, $\lambda = 0.3$, $\rho = 0.5$): $L^* = 0.3$, expected deadweight 0.045. No answer in the book.

10. News about prospects (Tirole 15.2)

(i) NPV $= \int_{y^*}^1 y\,dG - [1-G(y^*)]J - I$; investors' net $= \int_{y^*}^1 y\,dG - [1-G(y^*)](J+\mathcal{R}) - (I-A)$. **(ii)** $A \ge A_0^*$: $y^* = J$; $A_1^* < A < A_0^*$: smallest $y^* \in (J, J+\mathcal{R})$ with zero net income; $A < A_1^*$: no investment. For $A > A_1^*$, $y \in [y^*, J+\mathcal{R})$ cannot be refinanced on the market: arrange liquidity at date 0. **(iii)** Yes: aggregate net pledgeable income $= I - A > 0$ (if pooled). **(iv)** $E_\theta[\int_{y^*(\theta)}^1 (y - J - \mathcal{R})dG(y \mid \theta)] \ge I - A$; FOC $y^*(\theta) = J + \frac{\mu}{1+\mu}\mathcal{R}$ for all θ; outside liquidity $\ge$ the worst-state shortfall $-\min_\theta \int_{y^*}^1 (y - J - \mathcal{R})dG$; if scarce, $q > 1$, more interruption, less investment. Example (uniform, $J = 0.3$, $\mathcal{R} = 0.2$, $I = 0.5$, $A = 0.385$): $y^* = 0.358579$; two states ($U[0,1]$, $U[0,0.6]$, $A = 0.45$): $y^* = 0.341886$, $L \ge 0.0125$. Matches the book (pp. 620–621).

11. Market conditions and credit rationing (Tirole 13.3)

(i) $I^{\mathrm{FB}} = (\alpha p_H P)^{1/(1-\alpha)}$; $\mathcal{P}(I^{\mathrm{FB}}) - I^{\mathrm{FB}} = [\frac{1-\alpha}{\alpha} - \frac{p_H B}{\Delta p}]I^{\mathrm{FB}}$, so $I^{\mathrm{SB}} = I^{\mathrm{FB}}$ iff $P \ge P_0$, $I^{\mathrm{FB}}(P_0) = (K-A)/[\frac{1-\alpha}{\alpha} - \frac{p_H B}{\Delta p}]$; below, $p_H(PI^\alpha - BI/\Delta p) = I + K - A$. **(ii)** At P_0^-: $\frac{dI^{\mathrm{SB}}/dP}{dI^{\mathrm{FB}}/dP} = \frac{1-\alpha}{\alpha\, p_H B/\Delta p} > 1$. **(iii)** $P = P(p_H[R(I) + R(I_0)])$: partial crowding out, stronger where I^{SB} is steeper; total output

rises. Example ($\alpha = 0.5$, $p_H = 0.8$, $B/\Delta p = 0.5$, $K = 0.5$, $A = 0.1$): $P_0 = 2.041241$, slopes 1.632993 vs. 0.653197. Matches the book (pp. 617–618).

12. Downsizing and aggregate liquidity (Tirole 15.1)

(i) $p_H R'(I) = \frac{1}{1-\lambda} + \frac{\mu}{1+\mu}\frac{p_H B}{\Delta p}$, $p_H R'(J) = \rho + \frac{\mu}{1+\mu}\frac{p_H B}{\Delta p} \Rightarrow J < I$ iff $\rho > 1/(1-\lambda)$. **(ii)** $L^* = \rho J^* - p_H[R(J^*) - BJ^*/\Delta p]$ (book: ρJ^*). If $L^S < L^*$: $q > 1$, $p_H R'(J) = \rho + \frac{p_H B}{\Delta p}[1 - \frac{\lambda}{(1+\mu)(\lambda+q-1)}]$, so J falls; premium from breakeven with $-(q-1)L$, the two FOCs and $L = L^S$. **(iii)** $V = (1-\lambda)\mathcal{P}(I) + \lambda\mathcal{P}(J) > \lambda\rho J$: enough if pooled. Index suffices iff $(1-\lambda)\mathcal{P}(I) \geq (1+\lambda)[\rho J - \mathcal{P}(J)]$; otherwise credit lines. Example ($R = 2X^{0.8}$, $p_H = 0.9$, $B/\Delta p = 1$, $\lambda = 0.3$, $A = 0.2$, $\rho = 1.6$): $I^* = 0.643244$, $J^* = 0.383471$, $L^* = 0.122577$. Matches the book (pp. 619–620) except the L^* convention.

13. Liquid and illiquid assets are complements (Tirole 15.4)

$x = 0$ unprofitable $\Rightarrow$ investing firms set $x = 1$. $\bar{\bar{q}} = 1 + \frac{\rho_1 - 1 - \bar{\rho}}{\rho_H - \rho_0}$. $L^S \leq \underline{L}^S = \frac{(\rho_H - \rho_0)A}{\rho_1 - \rho_0}$: $q = \bar{\bar{q}}$, $I = L^S/(\rho_H - \rho_0)$; up to $\bar{L}^S = \frac{(\rho_H - \rho_0)A}{1+\bar{\rho}-\rho_0}$: $I = L^S/(\rho_H - \rho_0)$, $q \downarrow$; beyond: $q = 1$, $I = A/(1 + \bar{\rho} - \rho_0)$. I nondecreasing in L^S. Example ($\lambda = 0.5$, $\rho_L = 0.2$, $\rho_0 = 0.6$, $\rho_H = 1$, $\rho_1 = 2$, $A = 1$): $\bar{\bar{q}} = 2$, $\underline{L}^S = 0.285714$, $\bar{L}^S = 0.4$; $L^S = 0.35$: $I = 0.875$, $q = 1.357143$. Matches the book (p. 621).

14. Dynamics of income inequality (Tirole 13.2)

(i) $c = (1-a)y$, $\mathcal{L} = ay$, $U = y$. **(ii)** $I_t = A_t/(1 - \rho_0/(1+r))$, $y_t = (\rho_1 - \rho_0)I_t$, $A_{t+1} = \hat{A} + \beta A_t$, $\beta < 1$ $\Rightarrow A_t \to A_\infty$. **(iii)** $A_0^* = (1 - \rho_0/(1+r))\underline{I}$; lenders: $A_{t+1} = \hat{A} + a(1+r)A_t \to A_\infty^L = \hat{A}/[1 - a(1+r)]$. **(iv)** Equal: $\rho_1 = 1 + r$, $A_\infty = \hat{A}/(1 - a\rho_1)$, $\kappa = \rho_0/\rho_1$. Unequal: $\rho_1 > 1 + r$, $A_\infty^L < A_0^* \leq A_\infty$, $\kappa A_\infty^L = (1-\kappa)(I_\infty - A_\infty)$. Example ($a = 0.4$, $\hat{A} = 1$, $\rho_1 = 1.5$, $\rho_0 = 0.5$, $\underline{I} = 3.5$): equal ($r = 0.5$, $\kappa = 1/3$) and unequal ($r = 0.05$, $A_\infty = 4.230769$, $A_\infty^L = 1.724138 < A_0^* = 1.833333$, $\kappa = 0.690476$) coexist. Matches the book (p. 617).

15. Exchange-rate policy (Tirole 16.3)

(i) $d^* = p_H R_l^*$, $d = p_H R_l^S + (1 - p_H)R_l^F$. **(ii)** Pledgeable $= \rho_0 I + R_l^F/e \Rightarrow R_l^F = SI$, $I = A/\{1 - [(S/e) + \rho_0]\}$. **(iii)** Commitment: $v'(g^*) = e[1 - (\frac{S}{e} + p_H R - 1)\frac{dI}{de}\frac{de}{dg^*}] > e$ (underspending). Noncommitment: $v'(g^*) = e[1 - \frac{d}{e^2}\frac{de}{dg^*}] < e$ (overspending), more depreciation, less investment. **(iv)** g^* rises with nontradable debt d: each borrower's nontradable claims depreciate everyone's, lowering all borrowing capacities. Example ($u = 6\log c$, $v = \log g$, $p_H = 0.8$, $R = 2$, $\rho_0 = 0.4$, $S = 0.3$, $A = 1$, $\mathcal{R}^* = 2$): commitment $g^* = 0.508893$, $e = 1.504114$; noncommitment $g^* = 0.701486$, $e = 1.598417$. Matches the book (p. 623).